

Benson (Zhen Cong) Chen

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AI Engineer specializing in low-level GPU optimization for machine learning systems across mobile, desktop, and safety-critical environments. Llama.cpp contributor with cross-vendor GPU experience (Qualcomm, Nvidia, Apple) and a track record of measurable wins—1.5× Qwen3 inference speedup on mobile GPUs, 60% runtime reduction on GEMM kernels, and BitNet ternary-quantization support across heterogeneous GPUs.

Skills & Competencies

- Programming languages: C, C++, Python, Java
- ML API: Vulkan, CUDA, Metal, PyTorch
- LLM: Llama.cpp, Safety Critical Environment, Extreme Quantization, GPU Kernel Optimization, AI Inference, Edge Deployment

Work Experience

AI Research Engineer (Edge Inference) **Tether.io** **JAN 2026 – PRESENT**

- Develop and optimize kernels, quantization workflows, and fine-tuning systems for LLMs/SLMs across mobile and desktop GPUs.
- Optimized Qwen3 model on mobile Adreno 840 GPU to achieve 1.5× speedup compared to the previous version.
- Added support for BitNet heterogeneous GPU inference in llama.cpp by enabling ternary quantization with GPU acceleration using Vulkan and Metal.
- Authored a research paper documenting the implementation of BitNet LoRA fine-tuning on heterogeneous GPUs.

ML Software Developer (ML Inference) **Lynx Software** **JAN 2024 – DEC 2025**

- Benchmarked and optimized GEMM operations on an IMG GPU, reducing application runtime by 60% through memory tiling, register tiling, and thread coarsening; validated bottleneck-level gains using a custom profiling tool.
- Enabled efficient linear quantization for matrix multiplication to improve computational performance.
- Optimized GPU computation for safety-critical machine learning operations such as convolution.
- Implemented a testing library for the ML inference engine to thoroughly test each ML operation for various inputs using PyTorch.
- Designed and built a data processing pipeline for an internal LLM tool.
- Automated software testing by leveraging Jenkins and Python scripts.

ML Software Developer (ML Inference) **CoreAVI (Lynx Software)** **MAY 2023 – AUG 2023**

- Developed a custom machine learning engine for safety-critical environments in C.
- Implemented various machine learning operations using Vulkan compute shaders conforming to the NNEF specification.
- Designed debugging tools for developers working on the machine learning engine.
- Built demo projects running VGG16 and VGG18 vision models on the company's machine learning framework to perform object detection on Intel and AMD GPUs.

ML Software Developer (ML Inference) **CoreAVI (Lynx Software)** **SEPT 2023 – DEC 2023**

- Enhanced the machine learning engine to support additional ML architectures.
- Wrote Python scripts to automate the testing and build pipeline.
- Improved and debugged the machine learning inference engine.
- Refined the NNEF compiler to parse NNEF documents according to the NNEF specification.

Compiler Software Engineer **Huawei** **SEPT 2023 – DEC 2023**

- Generated and preprocessed training data for the AI compiler used to predict the best optimization for LLVM IR.
- Created an API for an AI compiler ML model to perform predictions on LLVM IR using SSH and network sockets.

Education

THE UNIVERSITY OF TEXAS AT AUSTIN **PRESENT**

- Master of Computer Science, Candidate

- Bachelor of Computer Science (Co-operative Program), Dean's Honour